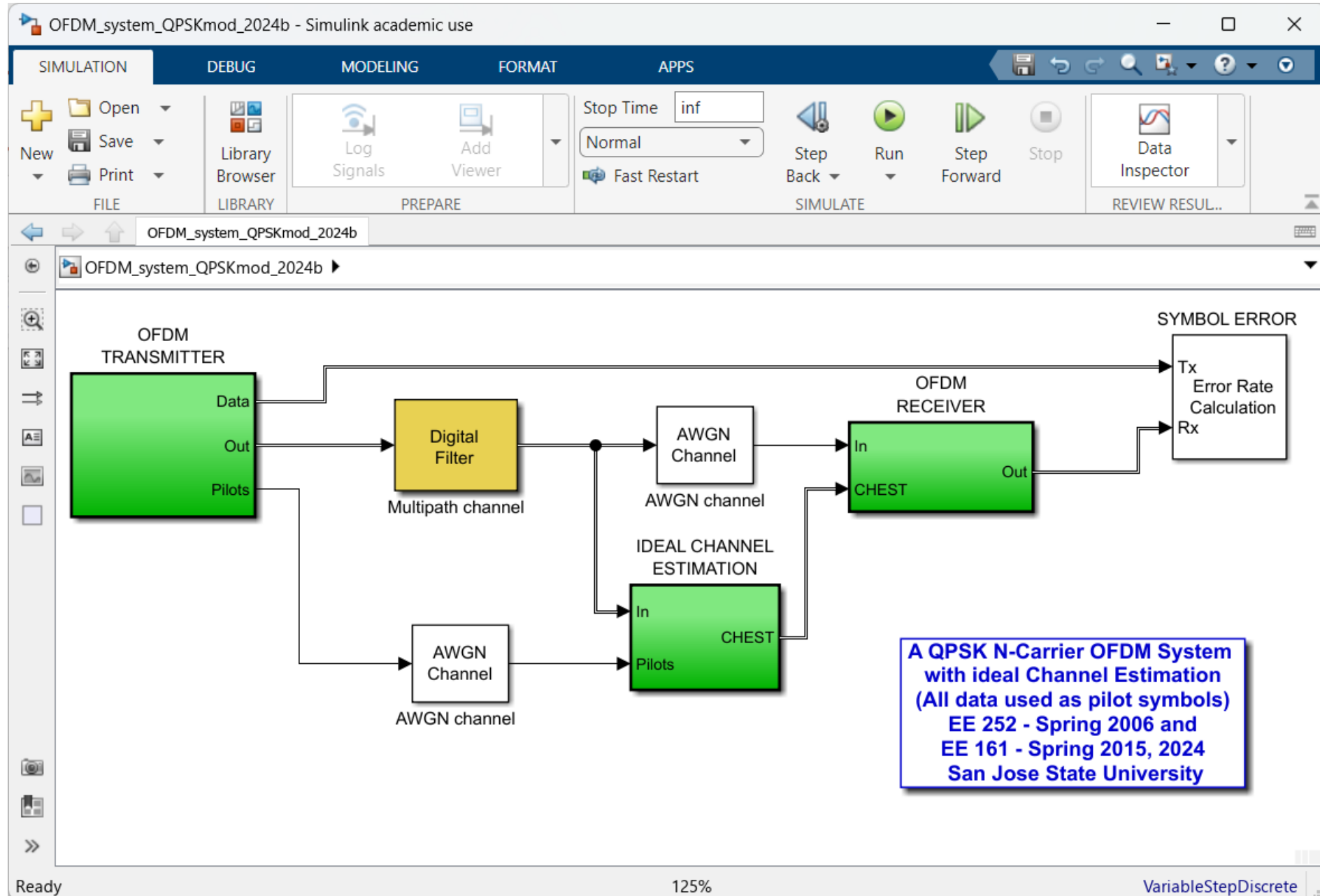


A note on error floors in OFDM

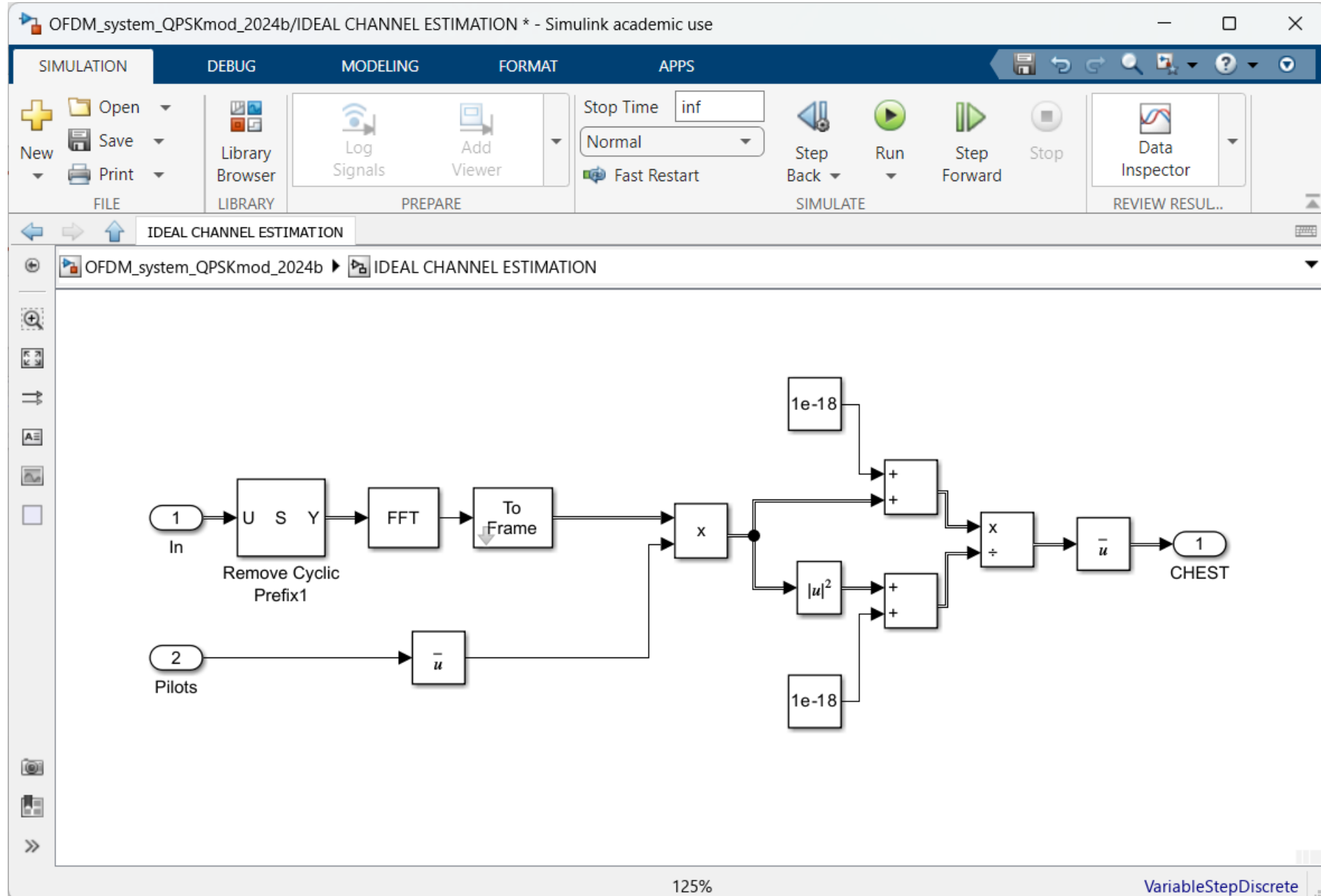
EE161

SJSU

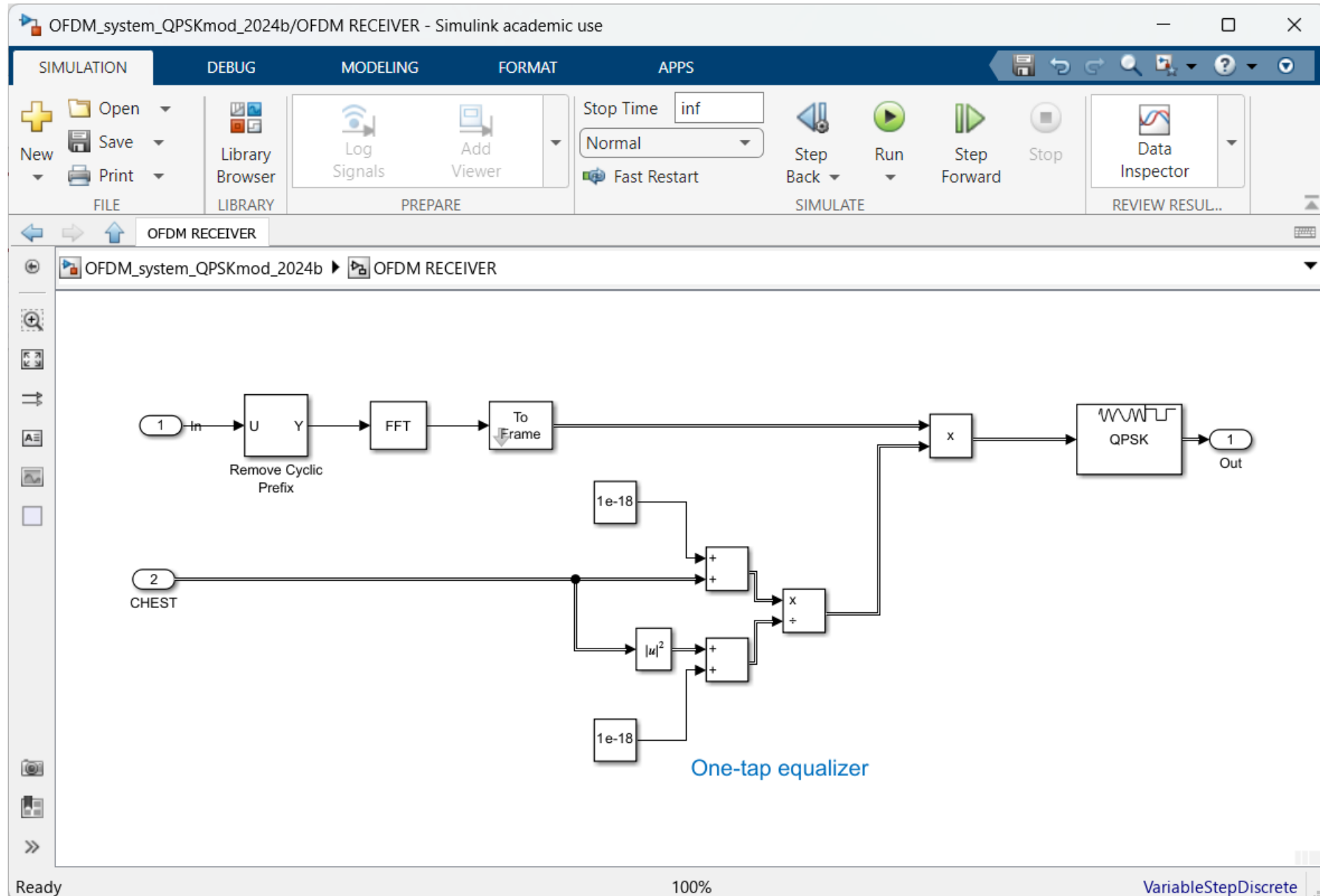
SJSU SAN JOSÉ STATE
UNIVERSITY



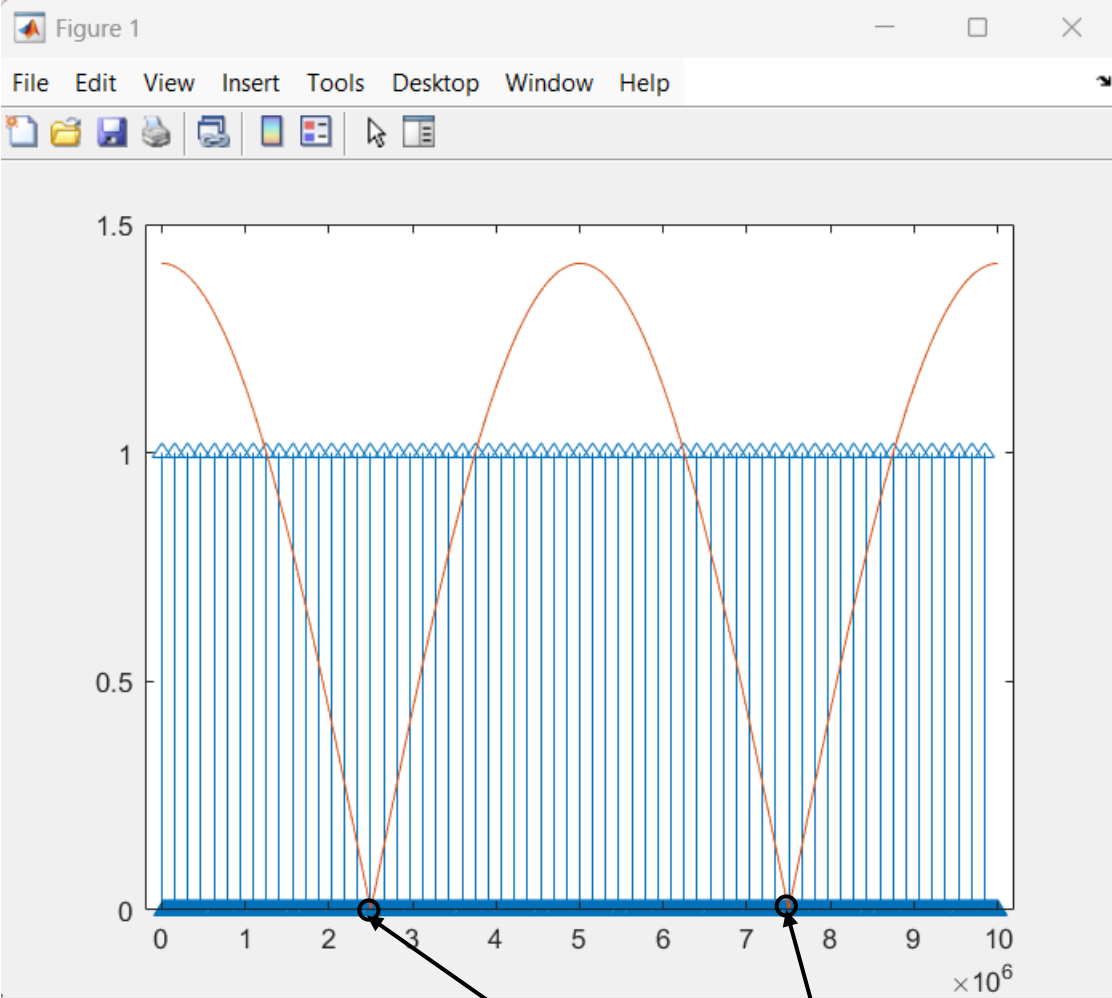
Channel estimation



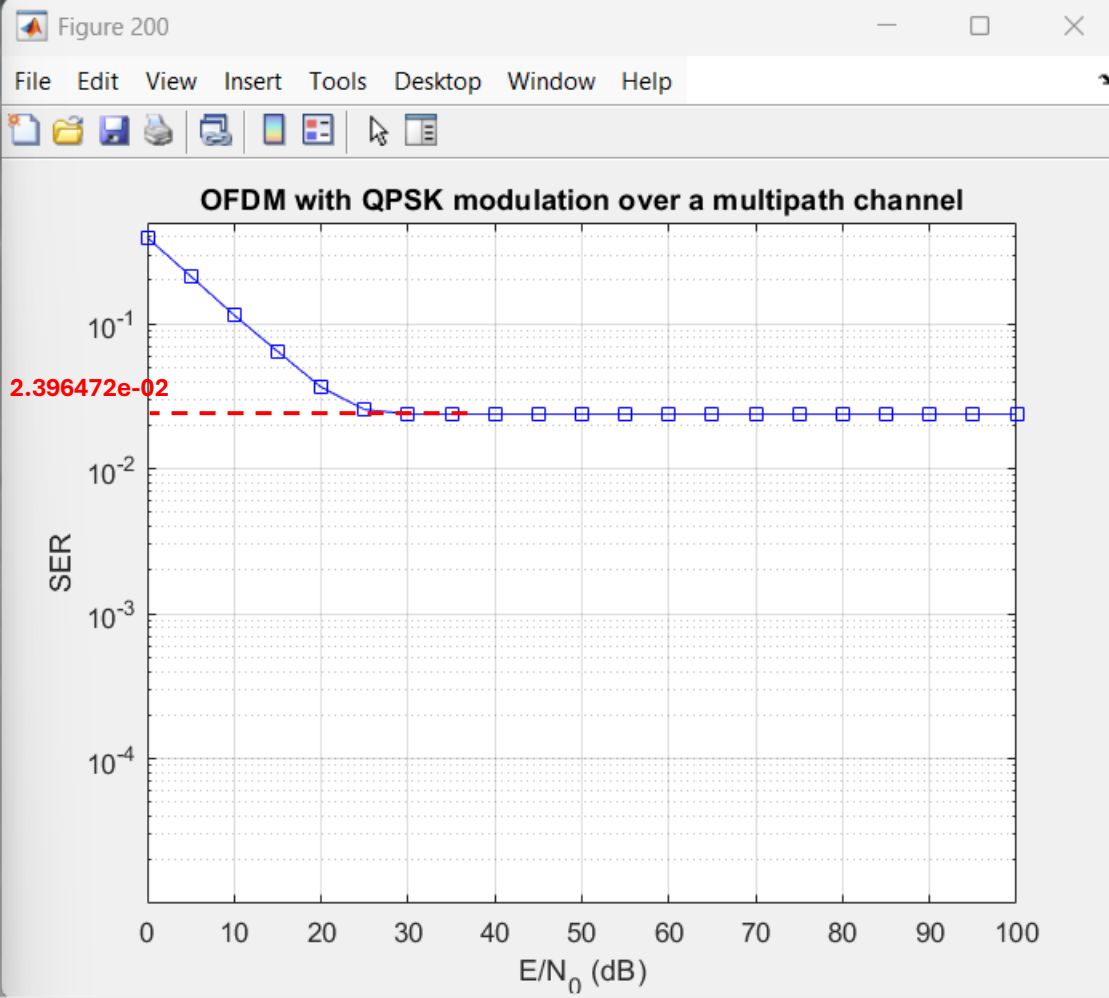
One-tap equalization



Error floor in OFDM. K=64. Multipath channel (symbol sample spaced): $[1 \ 0 \ 1]/\text{sqrt}(2)$



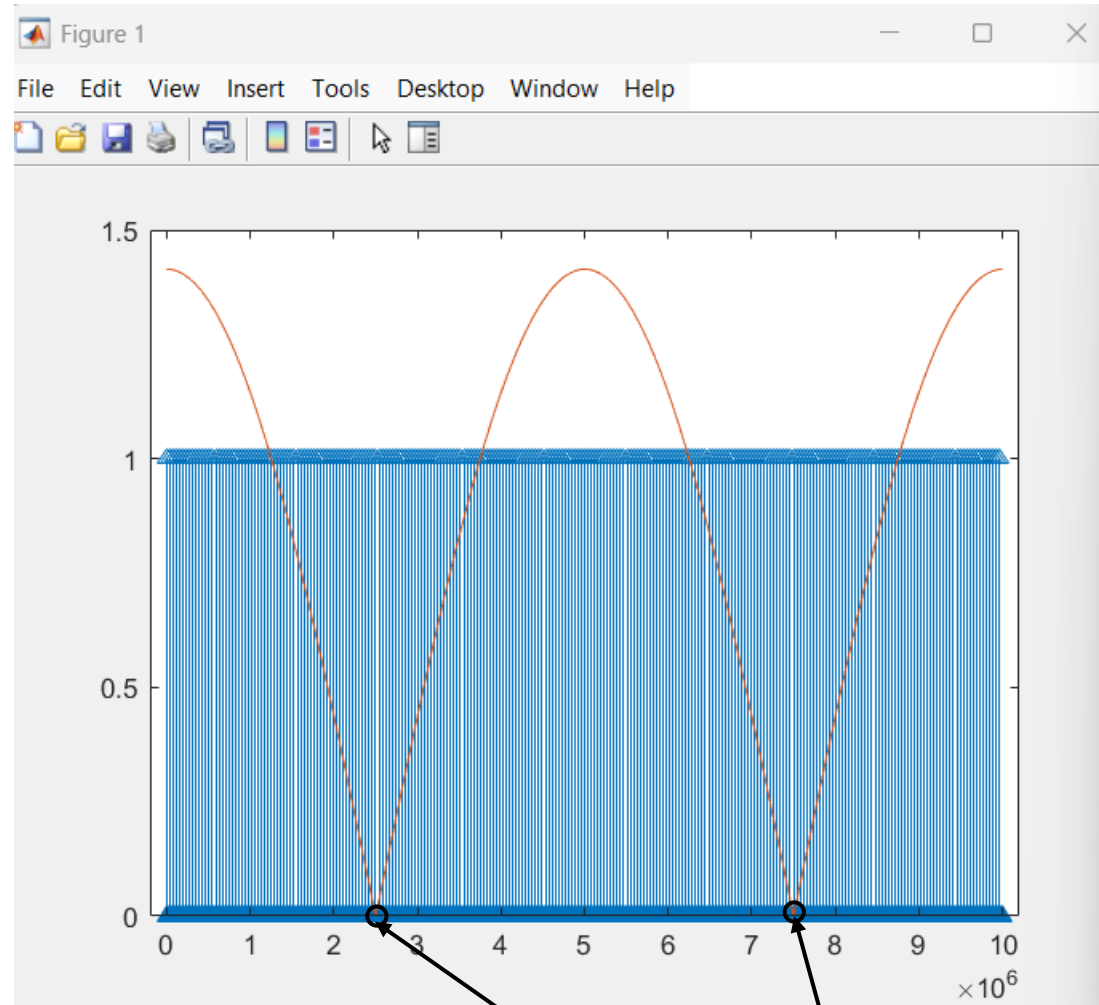
ofdm_multipath.m



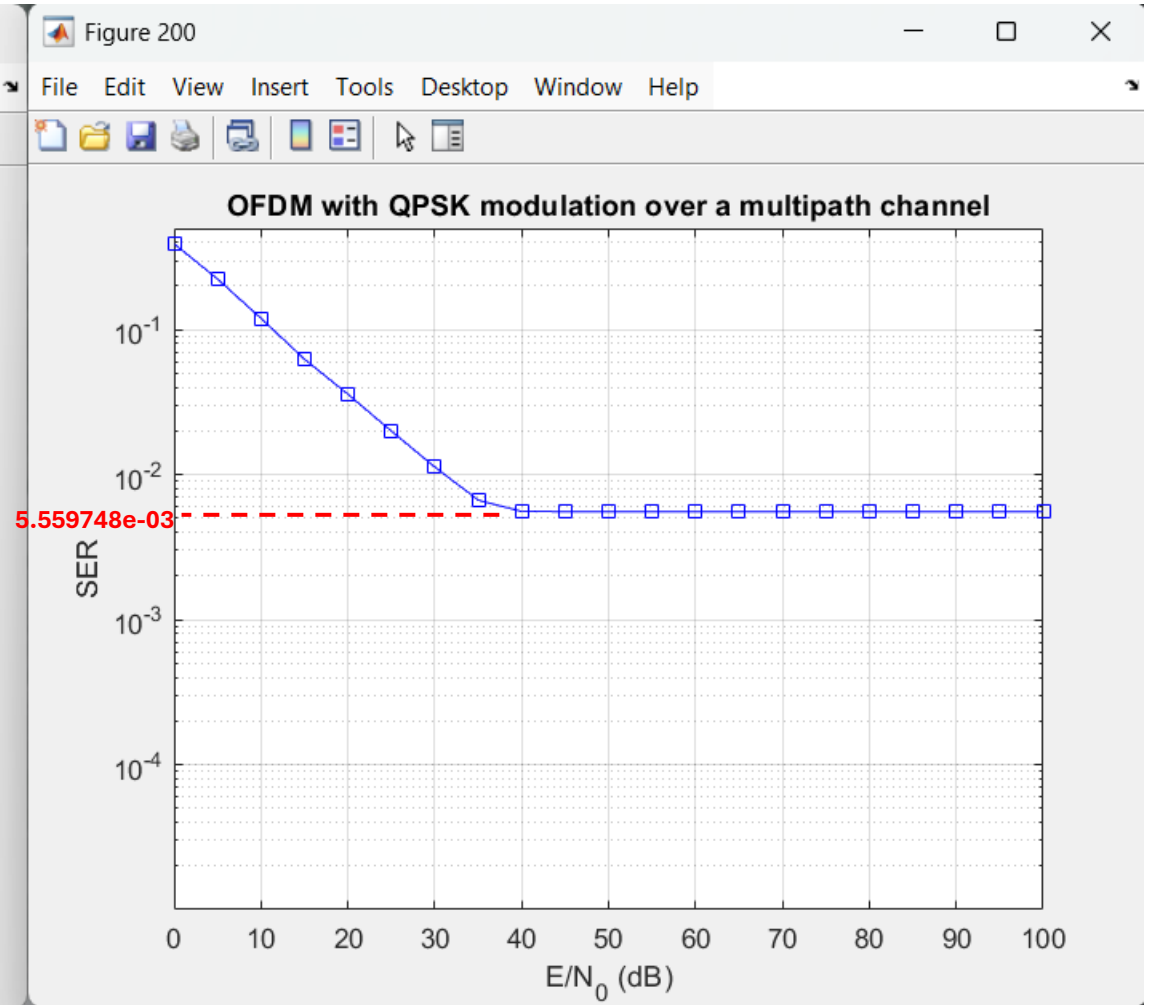
run_OFDM_2024b.m

$2 / 64 = 0.0312 \approx 2.396472e - 02$

Error floor in OFDM. K=256. Multipath channel (symbol sample spaced): $[1 \ 0 \ 1]/\sqrt{2}$



ofdm_multipath.m



run_OFDM_2024b.m

$$2 / 256 = 0.0078 \approx 5.559748e-03$$